

CTD

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2831 ±0.00355	0.4514 ±0.0179	0.3512 ±0.00834	0.3174 ±0.00336
Cramér $\ell_2^2 \downarrow$	0.02748 ±0.00115	0.02491 ±0.00213	0.03729 ±0.00147	0.0288 ±0.00138
CF-$\ell_2^2/\omega^2 \downarrow$	0.02115 ±0.000234	0.0201 ±0.00134	0.03157 ±0.000814	0.02292 ±0.000547

QTD

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2065 ±0.00985	0.4071 ±0.0249	0.3821 ±0.0348	0.2517 ±0.0181
Cramér $\ell_2^2 \downarrow$	0.0103 ±0.00105	0.01731 ±0.00323	0.0421 ±0.00913	0.0135 ±0.00348
CF-$\ell_2^2/\omega^2 \downarrow$	0.01003 ±0.00105	0.01655 ±0.00317	0.04171 ±0.00912	0.01299 ±0.00314

ϕ TD Dirac

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1362 ±0.00635	0.4517 ±0.0792	0.2314 ±0.046	0.179 ±0.0258
Cramér $\ell_2^2 \downarrow$	0.005086 ±0.000446	0.01702 ±0.00537	0.01096 ±0.00218	0.005498 ±0.00207
CF-$\ell_2^2/\omega^2 \downarrow$	0.004409 ±0.00046	0.01511 ±0.00531	0.009874 ±0.00193	0.004608 ±0.00174

ϕ TD Cat

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2887 ±0.00575	0.4229 ±0.041	0.3655 ±0.0333	0.3191 ±0.00605
Cramér $\ell_2^2 \downarrow$	0.02808 ±0.00137	0.02251 ±0.00181	0.03634 ±0.00212	0.02898 ±0.00152
CF-$\ell_2^2/\omega^2 \downarrow$	0.022 ±0.000503	0.01971 ±0.00242	0.03101 ±0.00177	0.02308 ±0.00063

ϕ TD Quant

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1367 ±0.0121	0.3889 ±0.0374	0.2066 ±0.0236	0.1635 ±0.0178
Cramér $\ell_2^2 \downarrow$	0.0046 ±0.000981	0.01071 ±0.00102	0.008993 ±0.00242	0.004491 ±0.00151
CF-$\ell_2^2/\omega^2 \downarrow$	0.004064 ±0.00102	0.00975 ±0.00162	0.008469 ±0.00236	0.003996 ±0.00155